

Ex1

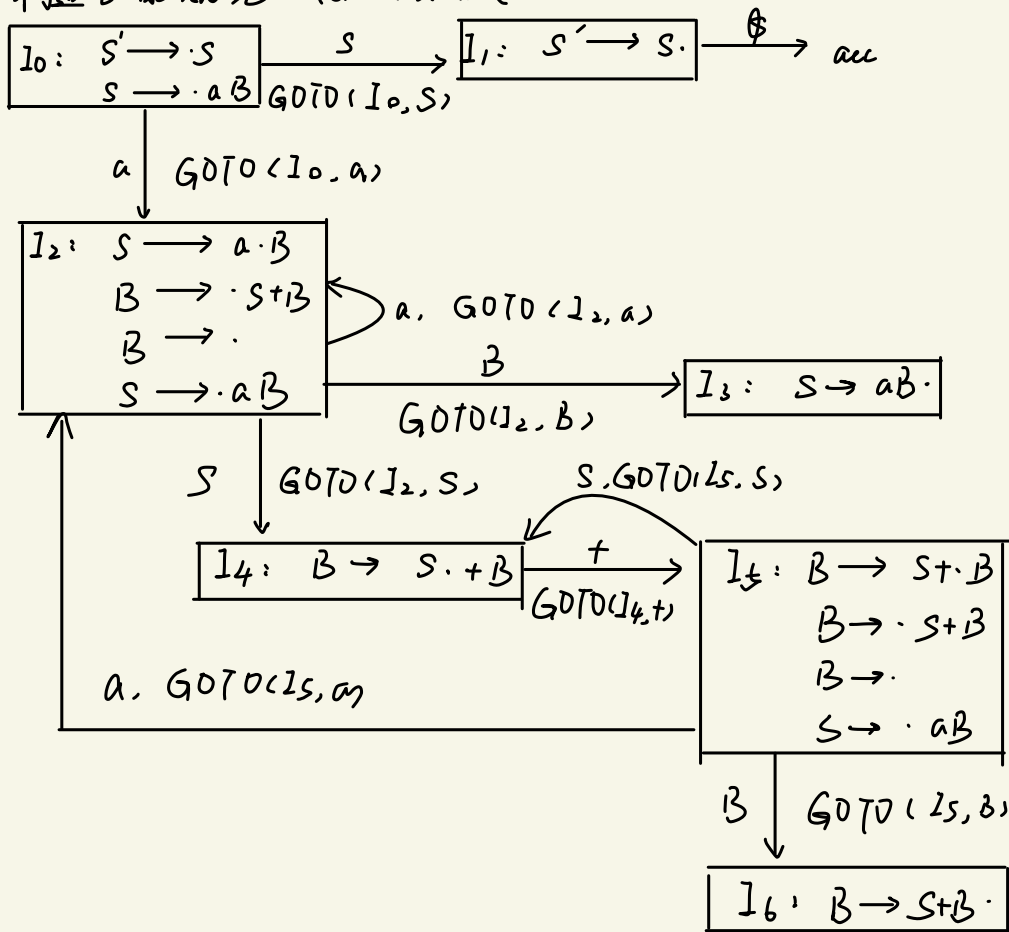
- 1, $S \rightarrow aB$
- 2, $B \rightarrow S+B$
- 3, $B \rightarrow \epsilon$

1. ① 构造增广文法 S'

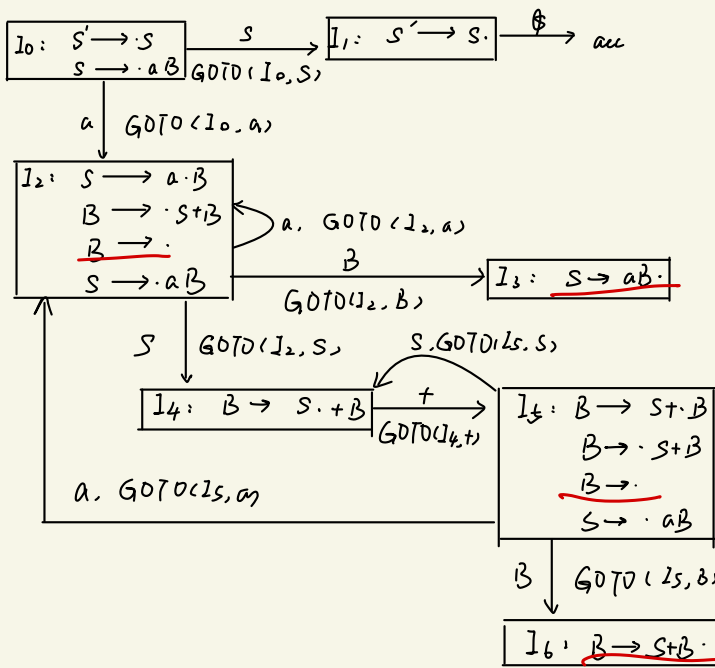
$$\begin{aligned} S' &\rightarrow S \\ S &\rightarrow aB \\ B &\rightarrow S+B \mid \epsilon \end{aligned}$$

$\{S, a, B, +, \epsilon, \$\}$
仅用于 r
 Δ

② 构造 S' 的规范 LR(0) 项集族



③ ACTION & ④ GOTO



$FOLLOW(S) = \{+, \$\}$

$FOLLOW(B) = \{+, \$\}$

状态	ACTION			GOTO	
	a	+	\$	S	B
0	S2			1	
1			acc		
2	S2	r3	r3	4	3
3		r1	r1		
4		S5			
5	S2	r3	r3	4	6
6		r2	r2		

2. Yes

3.

STATE	ACTION			GOTO	
	a	+	\$	S	B
0	S2			1	
1			acc		
2	S2	r3	r3	4	3
3		r1	r1		
4		S5			
5	S2	r3	r3	4	6
6		r2	r2		

1, $S \rightarrow aB$

2, $B \rightarrow S+B$

3, $B \rightarrow \epsilon$

Stack	Symbol	Input	ACTION
0		aaaat++\$	S2
02	a	aaaat++\$	S2
022	aa	aaaat++\$	S2
0222	aaa	aaaat++\$	S2
02222	aaaa	aaaat++\$	S2
022222	aaaaa	aaaat++\$	r3 ← pop E = 0
0222223	aaaaaB	aaaat++\$	r1
022224	aaaaS	aaaat++\$	S5
0222245	aaaS+	aaaat++\$	r3
02222456	aaaS+B	aaaat++\$	r2
02223	aaaB	aaaat++\$	r1
0224	aaS	aaaat++\$	S5
02245	aaS+	aaaat++\$	r3
022456	aaS+B	aaaat++\$	r2
0223	aaB	aaaat++\$	r1
024	aS	aaaat++\$	S5
0245	aS+	aaaat++\$	r3
02456	aS+B	aaaat++\$	r2
023	aB	aaaat++\$	r1
01	S	aaaat++\$	acc
	S'		

Ex 2.

(0) $S' \rightarrow S$

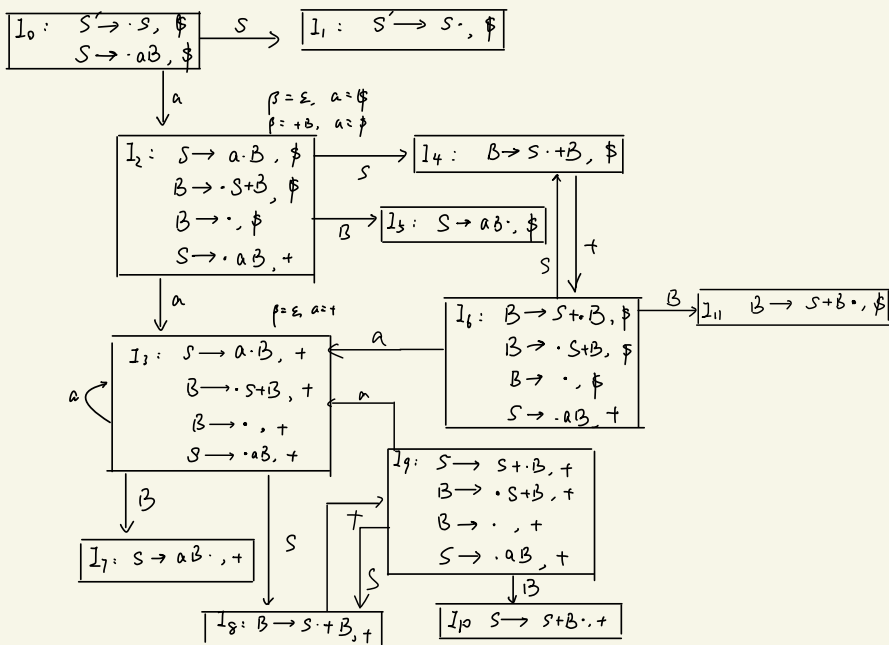
$[A \rightarrow \alpha \cdot B \beta, a]$

(1) $S \rightarrow aB$

$[LR(0) \text{ item, FIRST}(\beta a)]$

(2) $B \rightarrow S+B$

(3) $B \rightarrow \epsilon$ $\beta = \epsilon, a = \phi$



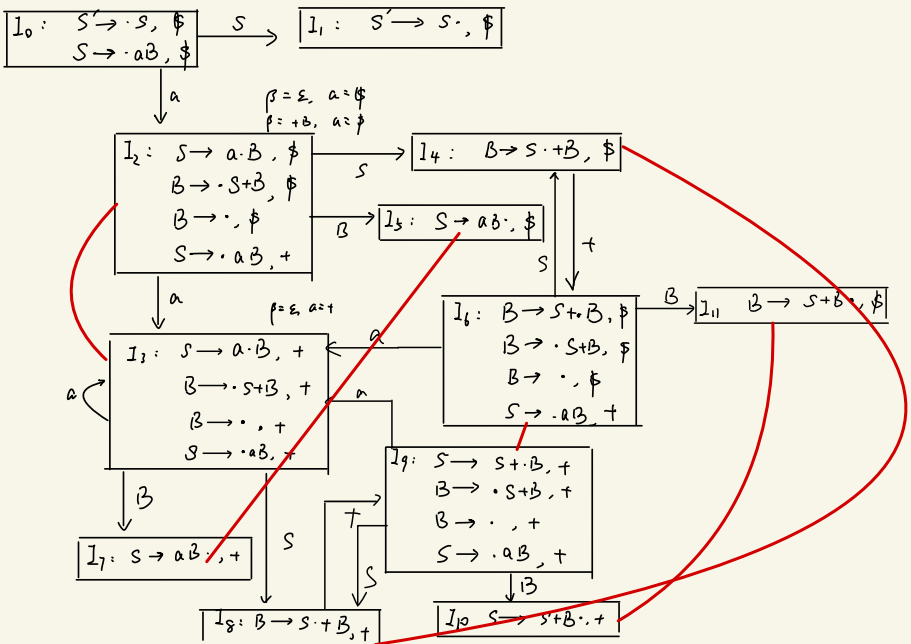
状态	ACTION			GOTO	
	a	+	ϕ	S	B
0	s2			1	
1			acc		
2	s3	r2	r3	4	5
3	s3	r3	r2	8	7
4		s6			
5		r1	r1		
6	s3	r2	r3	4	11
7		r1	r2		
8		s9			
9	s3	r3	r2	8	10
10		r2	r2		
11		r2	r2		

状态	ACTION			GOTO	
	a	+	\$	S	B
0	s ₂			1	
1			acc		
2	s ₃	r ₃	r ₃	4	5
3	s ₃	r ₃	r ₃	8	7
4		s ₆			
5		r ₁	r ₁		
6	s ₃	r ₃	r ₃	4	11
7		r ₁	r ₁		
8		s ₉			
9	s ₃	r ₃	r ₃	8	10
10		r ₂	r ₂		
11		r ₂	r ₂		

Stack	Symbol	Input	ACTION
0		aaaa++++\$	s ₂
0 2	a	aaa++++\$	s ₃
0 2 3	aa	aa++++\$	s ₃
0 2 3 3	aaa	a++++\$	s ₃
0 2 3 3 3	aaaa	++++\$	r ₃
0 2 3 3 7	aaaaB	++++\$	r ₁
0 2 3 3 8	aaaS	++++\$	s ₉
0 2 3 3 8 9	aaaS+	++\$	r ₃
0 2 3 3 8 9 10	aaaS+B	++\$	r ₂
0 2 3 3 7	aaaB	++\$	r ₁
0 2 3 8	aaS	++\$	s ₉
0 2 3 8 9	aaS+	+\$	r ₃
0 2 3 8 9 10	aaS+B	+\$	r ₂
0 2 3 7	aaB	+\$	r ₁
0 2 4	aS	+\$	s ₆
0 2 4 6	aS+	\$	r ₃
0 2 4 6 11	aS+B	\$	r ₂
0 2 5	aB	\$	r ₁
0 1	S	\$	acc

Ex 3.

$\beta = \epsilon, a = \beta$



状态	ACTION			GOTO	
	a	+	\$	S	B
0	s ₂			1	
1			acc		
2	s ₃	r ₃	r ₃	4	5
3	s ₃	r ₃	r ₃	8	7
4		s ₆			
5		r ₁	r ₁		
6	s ₃	r ₃	r ₃	4	11
7		r ₁	r ₁		
8		s ₉			
9	s ₃	r ₃	r ₃	8	10
10		r ₂	r ₂		
11		r ₂	r ₂		

状态	ACTION			GOTO	
	a	+	\$	S	B
0	s ₂₃			1	
1			acc		
23	s ₂₃	r ₃	r ₃	48	57
48		s ₆₉			
57		r ₁	r ₁		
69	s ₂₃	r ₃	r ₃	48	1011
1011		r ₂	r ₂		

Stack	Symbol	Input	ACTION
0		aaaa++t\$	S ₂₃
0 23	a	aaa++t\$	S ₂₃
0 23 23	a	aa++t\$	S ₂₃
0 23 23 23	a	a++t\$	S ₂₃
0 23 23 23 23	a	++t\$	r ₃
0 23 23 23 23 57	a	++t\$	r ₁
0 23 23 23 48	a	++t\$	S ₆₉
0 23 23 23 48 69	a	++t\$	r ₃
0 23 23 23 48 69 1011	a	++t\$	r ₂
0 23 23 23 57	a	++t\$	r ₁
0 23 23 48	a	++t\$	S ₆₉
0 23 23 48 69	a	++t\$	r ₃
0 23 23 48 69 1011	a	++t\$	r ₂
0 23 23 57	a	++t\$	r ₁
0 23 48	a	++t\$	S ₆₉
0 23 48 69	a	++t\$	r ₃
0 23 48 69 1011	a	++t\$	r ₂
0 23 57	a	++t\$	r ₁
0 1	S	\$	acc